

Quantum Systems Analysis

3D Simple Systems: Spherical Harmonics and Central Distributions

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Spherical Harmonics and Function Expansion on the Sphere

Computational Routine for $Y_\ell^m(\theta, \phi)$

To begin, we implement a routine capable of evaluating the spherical harmonics $Y_\ell^m(\theta, \phi)$ for any pair of indices (ℓ, m) and over two-dimensional grids of the angular variables (θ, ϕ) . The starting point is the analytical definition that arises from separating variables in the Laplacian under spherical coordinates.

The normalized spherical harmonics are defined as

$$Y_\ell^m(\theta, \phi) = \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - |m|)!}{(\ell + |m|)!}} P_\ell^{|m|}(\cos \theta) e^{im\phi}, \quad (1)$$

where $P_\ell^{|m|}(x)$ is the associated Legendre polynomial. This definition already incorporates the normalization factor and the Condon–Shortley convention $(-1)^m$, which is included by default in MATLAB’s `legendre` function.

The routine is implemented in fully vectorized form, meaning all operations are applied simultaneously to every element of the angular grids θ and ϕ . To evaluate Eq. (1) numerically, the routine follows these steps:

1. **Input.** The function receives the indices (ℓ, m) and two arrays of identical size containing the values of θ and ϕ , which allows it to work correctly over two-dimensional grids and produce surface plots.
2. **Associated Legendre polynomials.** Given the indices, we compute $z = \cos \theta$ and call `legendre(1, z)`, which returns all associated polynomials $P_\ell^m(z)$ for $m = 0, \dots, \ell$ including the $(-1)^m$ phase. From this output, only the component at order $|m|$ is needed.
3. **Normalization factor.** The prefactor of Eq. (1) is computed directly as

$$N_{\ell, m} = \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - |m|)!}{(\ell + |m|)!}}.$$

4. **Azimuthal dependence.** The complex factor $e^{im\phi}$ is generated to capture the rotational symmetry in the azimuthal direction.
5. **Final assembly.** The prefactor, the associated Legendre polynomial, and the complex phase are multiplied element-wise to yield the spherical harmonic at each point of the grid.

Figures 1 and 2 show the real and imaginary parts of the spherical harmonic evaluated for $(\ell, m) = (4, 3)$.

Parte real del armónico

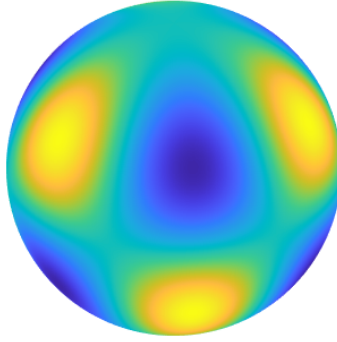


Figure 1: Real part of the spherical harmonic $Y_4^3(\theta, \phi)$.

Parte imaginaria del armónico

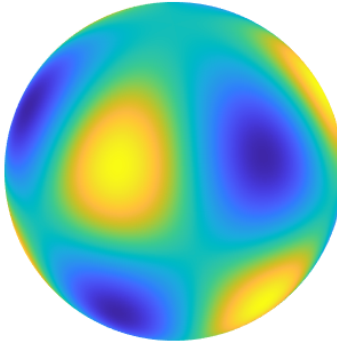


Figure 2: Imaginary part of the spherical harmonic $Y_4^3(\theta, \phi)$.

Expansion of $g(\theta, \phi)$ in Spherical Harmonics

Once the computational routine is in place, we expand an arbitrary angular function over the unit sphere in terms of this orthonormal basis. The function under study is

$$g(\theta, \phi) = A \theta \cos^3(\phi - 2\theta) (\pi - \theta)^2,$$

where A is a normalization constant ensuring that g has unit norm on the sphere.

2.1 Normalization Constant A

The normalization condition on the sphere is

$$\int_0^{2\pi} \int_0^\pi |g(\theta, \phi)|^2 \sin \theta \, d\theta \, d\phi = 1.$$

Substituting the definition of g ,

$$A^2 \int_0^{2\pi} \int_0^\pi \theta^2 \cos^6(\phi - 2\theta) (\pi - \theta)^4 \sin \theta d\theta d\phi = 1. \quad (2)$$

Since the only ϕ -dependent factor is $\cos^6(\phi - 2\theta)$, the inner integral can be isolated for each fixed θ :

$$I_\phi(\theta) = \int_0^{2\pi} \cos^6(\phi - 2\theta) d\phi.$$

Under the substitution $u = \phi - 2\theta$, this becomes

$$I_\phi(\theta) = \int_0^{2\pi} \cos^6 u du,$$

which is independent of θ because $\cos^6 u$ is 2π -periodic and any interval of length 2π yields the same result. Expanding using $\cos^2 u = (1 + \cos 2u)/2$,

$$\cos^6 u = \frac{5}{16} + \frac{15}{32} \cos 2u + \frac{3}{16} \cos 4u + \frac{1}{32} \cos 6u,$$

and integrating term by term over $[0, 2\pi]$, all oscillatory terms vanish and

$$\int_0^{2\pi} \cos^6 u du = \frac{5}{16}(2\pi) = \frac{5\pi}{8}.$$

Substituting back, the double integral reduces to a single integral in θ :

$$1 = A^2 \frac{5\pi}{8} \int_0^\pi \theta^2 (\pi - \theta)^4 \sin \theta d\theta. \quad (3)$$

This integral has no simple closed form and is evaluated numerically via the trapezoidal rule on a fine grid of $N_\theta = 5000$ points:

$$I_\theta \approx \sum_{i=1}^{N_\theta} \theta_i^2 (\pi - \theta_i)^4 \sin \theta_i \Delta\theta, \quad \Delta\theta = \frac{\pi}{N_\theta - 1}. \quad (4)$$

The resulting normalization constant is

$A \approx 0.148$

2.2 Surface Visualization of $g(\theta, \phi)$

With A determined, the function $g(\theta, \phi)$ is visualized on the unit sphere by constructing a grid in (θ, ϕ) , evaluating the function at each point, and converting to Cartesian coordinates. Figure 3 shows the result, revealing the regions where the function changes sign and its symmetry structure prior to the harmonic expansion.

Mapa superficial de $g(\theta, \phi)$

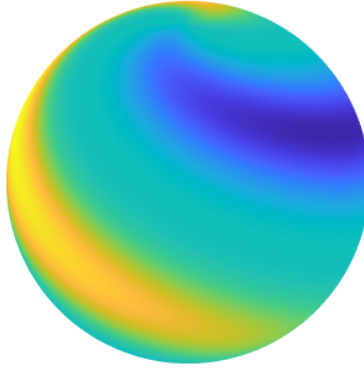


Figure 3: Surface visualization of $g(\theta, \phi)$ on the unit sphere.

2.3 Expansion Coefficients $C_{\ell m}$

The coefficients of the expansion in the orthonormal spherical harmonic basis are defined by

$$C_{\ell m} = \int_0^{2\pi} \int_0^\pi g(\theta, \phi) \overline{Y_\ell^m(\theta, \phi)} \sin \theta d\theta d\phi. \quad (5)$$

Since g does not admit a closed-form antiderivative, the integral is approximated numerically over the angular grid. For grid points θ_i and ϕ_j with spacings $\Delta\theta$ and $\Delta\phi$,

$$C_{\ell m} \approx \sum_{i=1}^{N_\theta} \sum_{j=1}^{N_\phi} g(\theta_i, \phi_j) \overline{Y_\ell^m(\theta_i, \phi_j)} \sin \theta_i \Delta\theta \Delta\phi.$$

2.4 Dominant Coefficients and Spectral Table

Table 1 lists the two largest-magnitude coefficients per degree ℓ , obtained for $\ell = 0, \dots, 7$.

ℓ	m	$\text{Re}(C_{\ell m})$	$\text{Im}(C_{\ell m})$	$ C_{\ell m} $
0	0	-2.99×10^{-3}	0	2.99×10^{-3}
1	1	3.78×10^{-1}	1.33×10^{-1}	4.01×10^{-1}
	-1	3.78×10^{-1}	-1.33×10^{-1}	4.01×10^{-1}
2	1	1.94×10^{-2}	4.29×10^{-1}	4.30×10^{-1}
	-1	1.94×10^{-2}	-4.29×10^{-1}	4.30×10^{-1}
3	1	-2.76×10^{-1}	1.21×10^{-1}	3.01×10^{-1}
	-1	-2.76×10^{-1}	-1.21×10^{-1}	3.01×10^{-1}
4	1	-9.93×10^{-2}	-8.57×10^{-2}	1.31×10^{-1}
	-1	-9.93×10^{-2}	8.57×10^{-2}	1.31×10^{-1}
5	3	-7.77×10^{-2}	-3.06×10^{-2}	8.35×10^{-2}
	-3	-7.77×10^{-2}	3.06×10^{-2}	8.35×10^{-2}
6	3	1.89×10^{-2}	-1.26×10^{-1}	1.27×10^{-1}
	-3	1.89×10^{-2}	1.26×10^{-1}	1.27×10^{-1}
7	3	1.30×10^{-1}	-1.49×10^{-2}	1.31×10^{-1}
	-3	1.30×10^{-1}	1.49×10^{-2}	1.31×10^{-1}

Table 1: Dominant coefficients $C_{\ell m}$ in the expansion of $g(\theta, \phi)$ (the two largest-magnitude values per degree ℓ are shown).

2.5 Amplitude Histogram

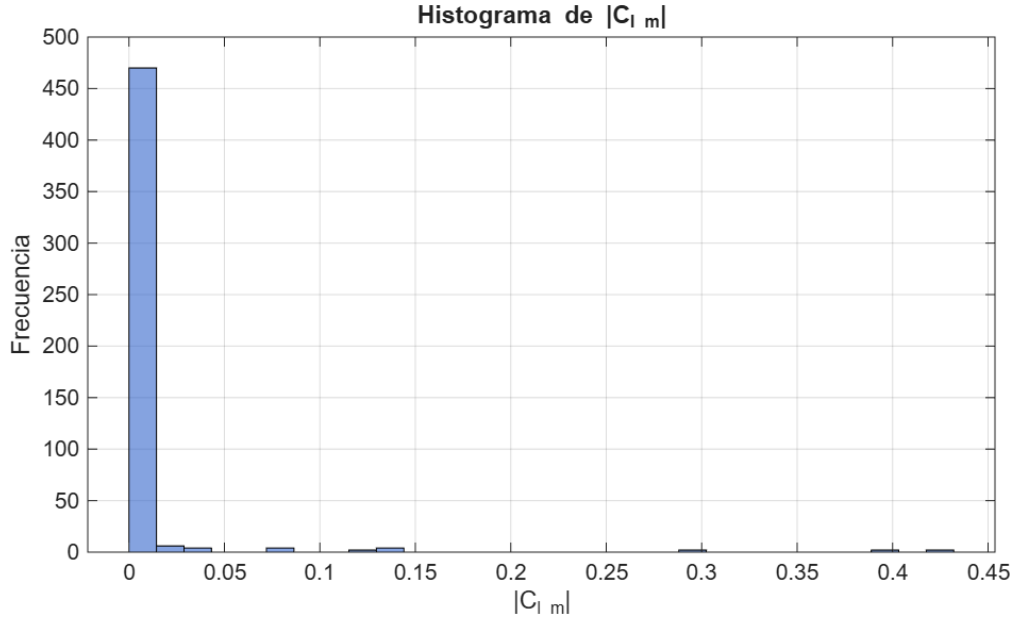


Figure 4: Histogram of the coefficient magnitudes $|C_{\ell m}|$.

The histogram shows that the vast majority of coefficients have magnitudes of order 10^{-4} to 10^{-3} , confirming that the function is well-localized and can be described adequately by a small number of harmonic modes. The dominant contributions — visible as outliers of order 10^{-1} — correspond to the modes with the most structural weight.

According to both the table and the histogram, the dominant mode corresponds to $\ell = 2$, $m = \pm 1$, with

$$|C_{2,\pm 1}| \approx 4.30 \times 10^{-1}.$$

2.6 Reconstruction Error and Truncation Criterion

Once the coefficients are computed, the reconstructed function

$$F(\theta, \phi) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} C_{\ell m} Y_{\ell}^m(\theta, \phi)$$

is compared against the original $g(\theta, \phi)$ via the pointwise relative error

$$\left| \frac{F(\theta, \phi) - g(\theta, \phi)}{g(\theta, \phi)} \right| < \text{tol}, \quad \text{tol} = 0.01. \quad (6)$$

A numerical subtlety arises: wherever $g(\theta, \phi) \approx 0$, the denominator in (6) becomes very small, producing artificially large relative errors even when the absolute reconstruction error is negligible. To address this, the relative error is computed only where $|g|$ exceeds a small

threshold, while the absolute error is reported separately in the near-zero regions. The results are

$$\text{Maximum relative error (Eq. (6))} = 1.8039 \times 10^6$$

$$\text{Maximum absolute error where } g \approx 0 = 1.7235 \times 10^{-2}$$

The large relative error is entirely due to the near-zero denominator and does not reflect poor reconstruction quality. The small absolute error in those regions confirms that $F(\theta, \phi)$ faithfully reproduces the global structure of $g(\theta, \phi)$. The 1% tolerance criterion of Eq. (6) cannot be applied strictly when g has nodes, since the criterion itself becomes numerically unstable in those regions.

2.7 Numerical Reconstruction on the Sphere

Función original $g(\theta, \phi)$

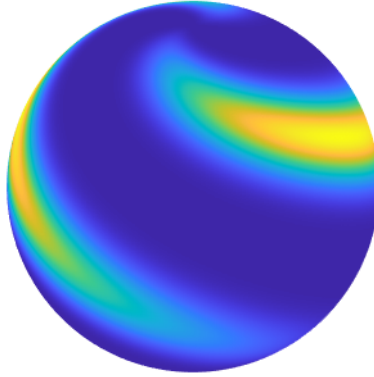


Figure 5: Original function $g(\theta, \phi)$ on the sphere.

Función reconstruida)

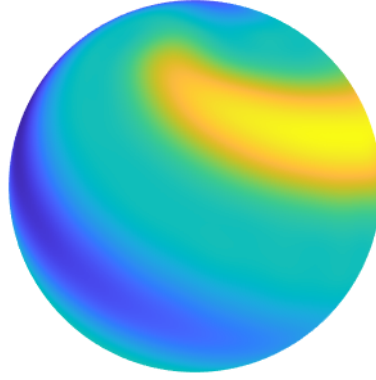


Figure 6: Reconstructed function $F(\theta, \phi)$ from the truncated spherical harmonic expansion.

The truncated expansion retains only low- and mid-order modes, discarding higher-order contributions that encode fine-scale variations. Despite this, the reconstruction clearly preserves the global structure of g , with only mild smoothing in localized regions — a direct consequence of the series truncation.

Visualization with the Hammer Cartographic Projection

Definition and Numerical Implementation

The Hammer projection maps functions defined on the unit sphere to a two-dimensional plane embedded in an ellipse. The values of the function are preserved; only the parametrization of the domain changes.

For an arbitrary function $f(\theta, \phi)$, the transformation is

$$x = \frac{\sqrt{8} \sin \theta \sin(\phi/2)}{\sqrt{1 + \sin \theta \cos(\phi/2)}}, \quad y = \frac{\sqrt{2} \cos \theta}{\sqrt{1 + \sin \theta \cos(\phi/2)}},$$

where $\theta \in [0, \pi]$ and $\phi \in (-\pi, \pi]$. The numerical implementation proceeds as follows:

1. **Input.** A grid (Θ, Φ) of colatitude and azimuthal angle values, together with a scalar field $G(\Theta, \Phi)$ already evaluated on that grid.
2. **Hammer coordinates.** The two-dimensional arrays $X(\Theta, \Phi)$ and $Y(\Theta, \Phi)$ are computed elementwise using the formulas above.
3. **Output.** The function returns the grids X and Y of the same size as G . Each point (X_{ij}, Y_{ij}) is associated with the field value G_{ij} .

$g(\theta, \phi)$ on the Hammer Ellipse

With $A \approx 0.148$, the function is

$$g(\theta, \phi) = 0.148 \theta \cos^3(\phi - 2\theta) (\pi - \theta)^2.$$

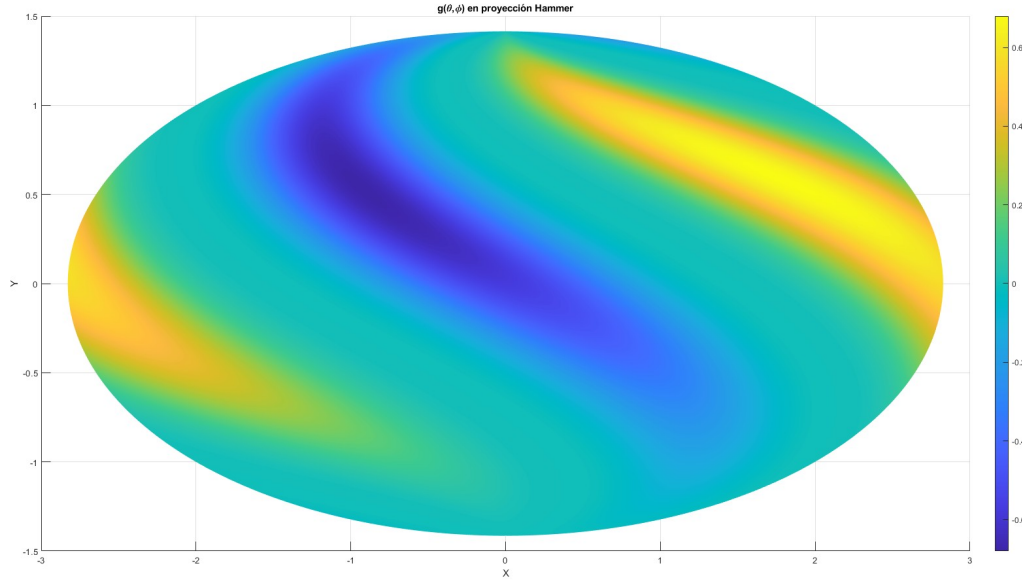


Figure 7: Hammer projection of $g(\theta, \phi)$.

The elliptical boundary in Figure 7 represents the entire spherical surface. Color bands extend across the ellipse, with yellow regions indicating the positive maxima of g and blue regions the minima. Intermediate tones correspond to values near zero.

Real Part of $Y_6^9(\theta, \phi)$ in Hammer

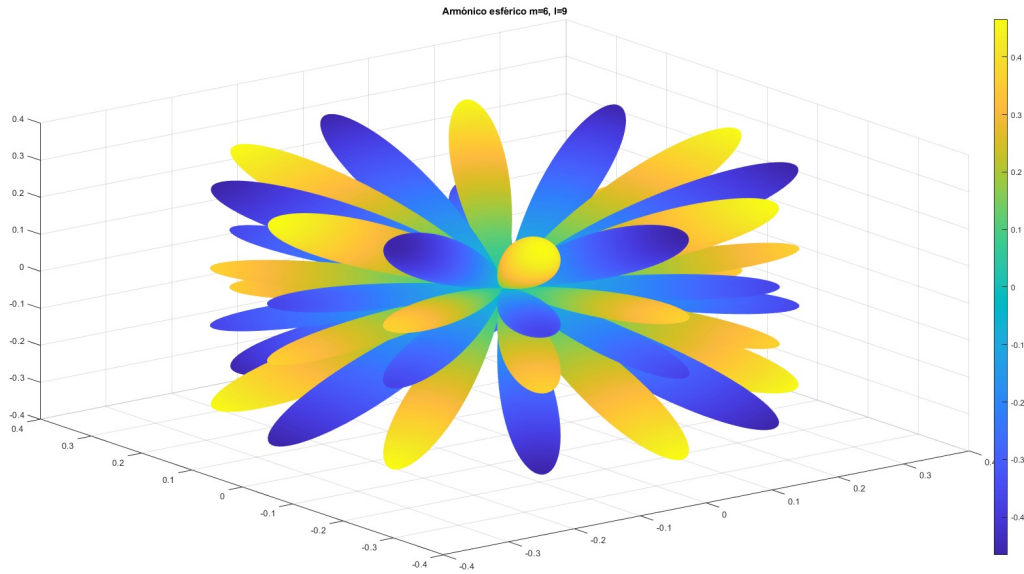


Figure 8: Real part of the spherical harmonic $Y_6^9(\theta, \phi)$ on the sphere.

This harmonic displays 12 lobes in the xy -plane and 4 lobes in height, alternating continuously between its minimum (blue) and maximum (yellow) values.

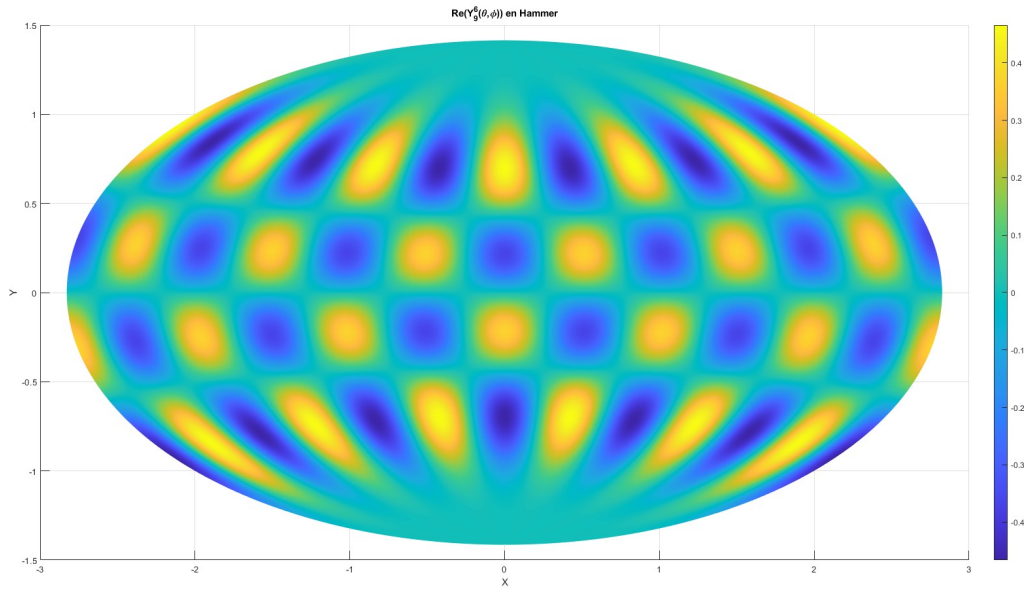


Figure 9: Hammer projection of $\text{Re}[Y_6^9(\theta, \phi)]$.

The Hammer projection preserves the alternating lobe pattern, distributing the sign changes across the full extent of the ellipse.

Real Part of the Spin Function $G_6^9(\theta, \phi)$ in Hammer

Under certain conditions, a spin state can be mapped to the surface of a sphere as

$$G_s^m(\theta, \phi) = \sqrt{\frac{2s+1}{4\pi} \frac{(2s)!}{(s+m)!(s-m)!}} \left(\cos \frac{\theta}{2}\right)^{s+m} \left(\sin \frac{\theta}{2}\right)^{s-m} e^{im\phi},$$

where $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ and $m = -s, -s+1, \dots, s$.

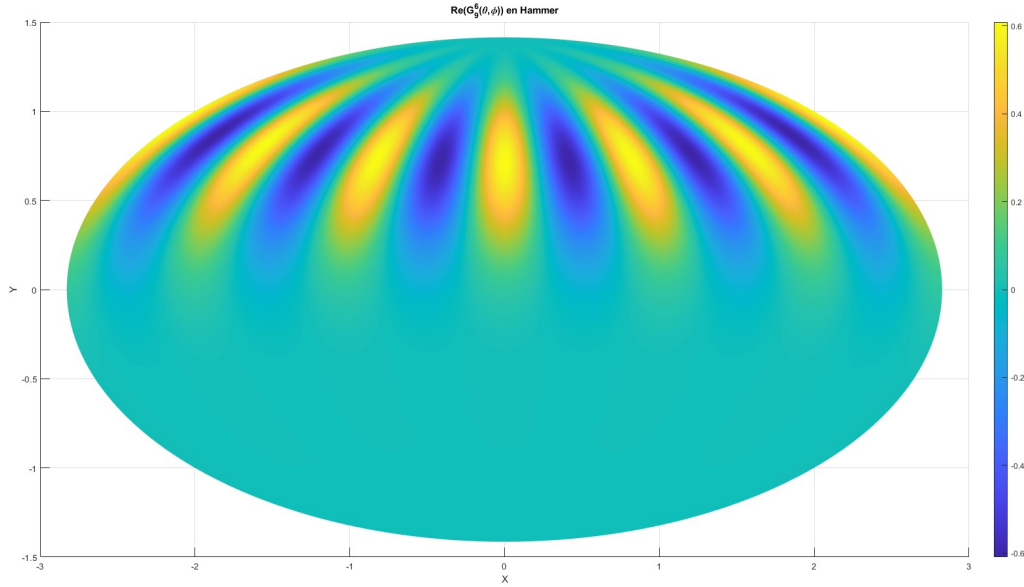


Figure 10: Hammer projection of $\text{Re}[G_9^6(\theta, \phi)]$.

Figure 10 shows that G_6^9 is concentrated primarily in the upper region of the ellipse, where a series of elongated lobes alternating between positive (yellow) and negative (blue) values are visible. The lower portion presents near-zero, nearly uniform values.

Probability Density $|G_{-3/2}^{5/2}(\theta, \phi)|^2$ in Hammer

The probability density of the spin state is $|G_s^m|^2$.

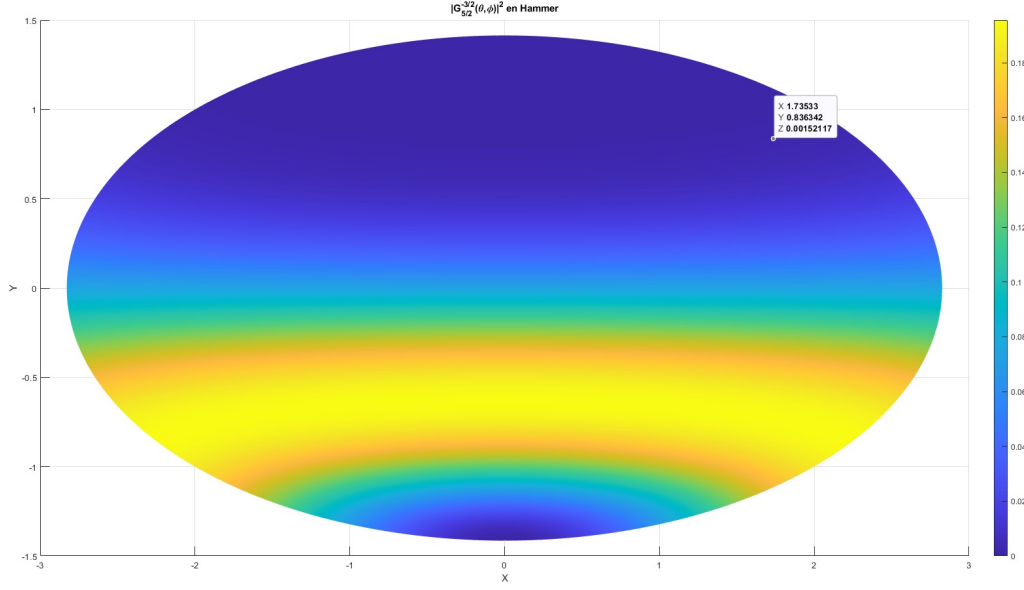


Figure 11: Probability density of $G_{-3/2}^{5/2}$ in the Hammer projection.

The density is non-uniform: the region near the lower part of the ellipse carries the highest values (yellow), while the upper zone is dominated by near-zero probabilities (blue). A gradual transition band separates both regimes.

Tennis Ball and Spherical Harmonic Expansion

Analytical Model of $T(\theta, \phi)$

A tennis ball is divided into two panels separated by a seam that wraps around the sphere. To model it, we construct a function that takes only the values $+1$ and -1 , representing the two panels.

The seam is modeled as a smooth deviation from the equator $\theta = \pi/2$:

$$\theta_{\text{seam}}(\phi) = \frac{\pi}{2} + \Delta\theta \sin(2\phi).$$

The parameter $\Delta\theta$ controls the amplitude of the undulation, and the factor $\sin(2\phi)$ introduces two complete oscillations per full azimuthal revolution, reproducing the characteristic shape of the seam. The panel function is then

$$T(\theta, \phi) = \begin{cases} +1, & \theta > \theta_{\text{seam}}(\phi), \\ -1, & \theta < \theta_{\text{seam}}(\phi). \end{cases}$$

This construction yields a piecewise-constant function with a single discontinuity along the seam curve.

Visualization on the Sphere and in the Hammer Projection

An angular grid with $\Delta\theta = \Delta\phi \approx 0.01$ rad is constructed over $\theta \in [0, \pi]$ and $\phi \in [-\pi, \pi]$. The seam curve $\theta_{\text{seam}}(\phi)$ is evaluated at each azimuthal point, and the two panels are assigned accordingly. The grid is then transformed to Cartesian coordinates and to the Hammer parametrization using

$$x_H = \frac{\sqrt{8} \sin \theta \sin(\phi/2)}{\sqrt{1 + \sin \theta \cos(\phi/2)}}, \quad y_H = \frac{\sqrt{2} \cos \theta}{\sqrt{1 + \sin \theta \cos(\phi/2)}}.$$



Figure 12: Tennis ball model $T(\theta, \phi)$ on the sphere.

$T(\theta, \phi)$ en la proyección de Hammer

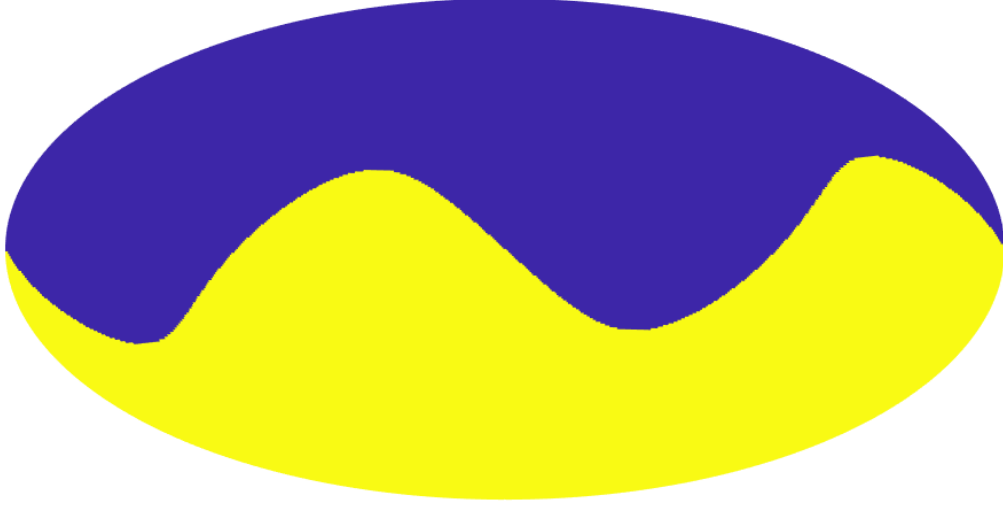


Figure 13: Hammer projection of $T(\theta, \phi)$.

The Hammer projection clearly shows the binary distribution of T , with the seam appearing as a sinusoidal horizontal band — consistent with the two-oscillation structure of $\sin(2\phi)$.

Harmonic Spectrum of $T(\theta, \phi)$

Since the spherical harmonics form an orthonormal basis on the sphere, T is expanded as

$$T(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} C_{\ell}^m Y_{\ell}^m(\theta, \phi),$$

with coefficients

$$C_{\ell}^m = \int_0^{2\pi} \int_0^{\pi} T(\theta, \phi) \overline{Y_{\ell}^m(\theta, \phi)} \sin \theta \, d\theta \, d\phi,$$

computed numerically using the same vectorized approach as for g , up to $\ell_{\max} = 15$.

ℓ	m	$\text{Re}(C_\ell^m)$	$\text{Im}(C_\ell^m)$	$ C_\ell^m $
1	0	-2.6799	0	2.6799
2	-2	-9.39×10^{-4}	-1.1507	1.1507
2	2	-9.39×10^{-4}	1.1507	1.1507
3	0	0.41456	0	0.41456
4	-2	6.64×10^{-4}	0.58431	0.58431
4	2	6.64×10^{-4}	-0.58431	0.58431
5	-4	-0.44565	3.58×10^{-4}	0.44565
5	0	0.13142	0	0.13142
6	-2	-4.58×10^{-4}	-0.16803	0.16803
6	2	-4.58×10^{-4}	0.16803	0.16803
7	-4	0.32442	-3.91×10^{-4}	0.32442
7	0	-0.20061	0	0.20061
8	-6	4.42×10^{-4}	0.26029	0.26029
8	6	4.42×10^{-4}	-0.26029	0.26029
9	-8	-0.089377	7.11×10^{-4}	0.08938
9	0	0.07746	0	0.07746
10	-6	-1.11×10^{-4}	-0.21853	0.21853
10	2	-3.18×10^{-6}	-0.12788	0.12788
11	-8	0.18036	-1.02×10^{-3}	0.18036
11	0	0.048384	0	0.048384
12	-10	-2.26×10^{-4}	0.076751	0.076751
12	2	-2.65×10^{-4}	0.053	0.053
13	-8	-0.16153	1.04×10^{-3}	0.16153
13	0	-0.082541	0	0.082541
14	-10	6.15×10^{-5}	-0.13838	0.13838
14	2	5.45×10^{-4}	0.035129	0.035133
15	-12	0.066904	-9.31×10^{-4}	0.066911
15	0	0.035485	0	0.035485

Table 2: Most representative coefficients C_ℓ^m in the spherical harmonic expansion of the tennis ball function $T(\theta, \phi)$.

Reconstruction of $T(\theta, \phi)$ on the Sphere

The reconstructed function is assembled by summing the contributions of all harmonics up to $\ell_{\max} = 15$, each weighted by its coefficient.

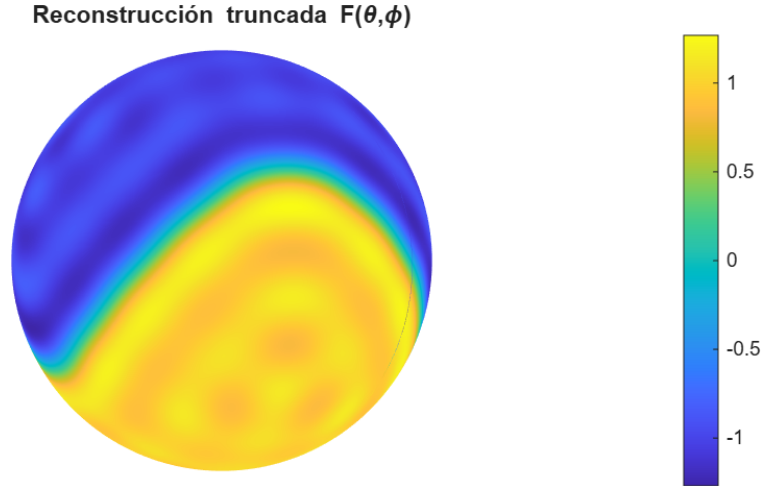


Figure 14: Truncated reconstruction of $T(\theta, \phi)$ on the sphere.

The general structure of the tennis ball — two panels separated by the seam — is correctly reproduced. However, the boundary between panels appears smoothed and exhibits oscillations around the seam, a direct consequence of truncating the series at finite $\ell_{\max}$.

Effect of the Discontinuity on the Harmonic Expansion

The function $T(\theta, \phi)$ has a single discontinuity along the seam separating the two panels. In the context of spherical harmonic expansions, such a jump discontinuity has a well-known consequence: the series converges slowly, requiring many high-order modes to resolve the sharp boundary. When the expansion is truncated at a finite $\ell_{\max}$, oscillations and overshoot appear near the discontinuity — the spherical analogue of the Gibbs phenomenon.

Free Particle on the Surface of a Sphere

Stationary States and Allowed Eigenenergies

Since the particle moves freely on the spherical surface of radius α , the potential vanishes and the Hamiltonian reduces to the kinetic energy:

$$H = \frac{p^2}{2m}, \quad p = -i\hbar\nabla.$$

Confined to the sphere, the radial coordinate is constant and the Laplacian reduces to its angular part:

$$\nabla^2 = \frac{1}{\alpha^2} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) = \frac{1}{\alpha^2} \nabla_{\Omega}^2.$$

The Hamiltonian therefore becomes

$$H = -\frac{\hbar^2}{2m\alpha^2} \nabla_{\Omega}^2.$$

Recalling the quantum mechanical angular momentum operator $\mathbf{L} = \frac{\hbar}{i} \mathbf{r} \times \nabla$, one can show that $L^2 = -\hbar^2 \nabla_\Omega^2$, so

$$H = \frac{L^2}{2m\alpha^2}.$$

The eigenfunctions of L^2 are the spherical harmonics:

$$L^2 Y_\ell^m(\theta, \phi) = \hbar^2 \ell(\ell + 1) Y_\ell^m(\theta, \phi).$$

Since the spherical harmonics are already normalized over angular coordinates, the stationary states are

$$\psi_{\ell m}(\theta, \phi) = \frac{1}{\alpha} Y_\ell^m(\theta, \phi),$$

with eigenenergies

$$E_\ell = \frac{\hbar^2 \ell(\ell + 1)}{2m\alpha^2},$$

depending only on ℓ , with degeneracy $2\ell + 1$ for each level. The quantum numbers run as $\ell = 0, 1, 2, \dots$ and $m = -\ell, -\ell + 1, \dots, \ell$.

Probability of Measuring $E = 3\hbar^2/(m\alpha^2)$

The initial state is the Gaussian wave packet

$$H(\theta, \phi; t = 0) = C \exp \left[- \left(\frac{\theta - \pi/3}{\pi/12} \right)^2 \right] \exp \left[- \left(\frac{\phi - \pi/2}{\pi/6} \right)^2 \right].$$

The normalization constant satisfies

$$\int_{-\pi}^{\pi} \int_0^{\pi} |H(\theta, \phi; 0)|^2 \sin \theta \, d\theta \, d\phi = 1,$$

which factorizes as $C^2 I_\phi I_\theta = 1$ due to the separable Gaussian form. Under the substitution $u = \sqrt{2}(\phi - \pi/2)/(\pi/6)$,

$$I_\phi = \frac{\pi}{6\sqrt{2}} \frac{\sqrt{\pi}}{2} \left[\operatorname{erf}(3\sqrt{2}) - \operatorname{erf}(-9\sqrt{2}) \right] \approx 0.6562,$$

while the polar integral is evaluated numerically:

$$I_\theta = \int_0^{\pi} \exp \left[-2 \left(\frac{\theta - \pi/3}{\pi/12} \right)^2 \right] \sin \theta \, d\theta \approx 0.2817.$$

Thus

$$C = \frac{1}{\sqrt{I_\phi I_\theta}} \approx 2.3257.$$

The energy $E = 3\hbar^2/(m\alpha^2)$ corresponds to $\ell = 2$, since

$$\frac{\hbar^2\ell(\ell+1)}{2m\alpha^2} = \frac{3\hbar^2}{m\alpha^2} \Rightarrow \ell(\ell+1) = 6 \Rightarrow \ell = 2.$$

The probability of measuring this energy is the sum of squared expansion coefficients over all m at $\ell = 2$:

$$c_{\ell m} = \int_{-\pi}^{\pi} \int_0^{\pi} \overline{Y_{\ell}^m(\theta, \phi)} H(\theta, \phi; 0) \sin \theta d\theta d\phi.$$

ℓ	m	$\text{Re}(c_{\ell m})$	$\text{Im}(c_{\ell m})$
0	0	0.240518	0
1	-1	2.72×10^{-6}	0.236978
1	0	0.197858	0
1	1	-2.72×10^{-6}	0.236978
2	-2	-0.187783	-4.24×10^{-7}
2	-1	2.80×10^{-6}	0.243585
2	0	-0.0672269	0
2	1	-2.80×10^{-6}	0.243585
2	2	-0.187783	4.24×10^{-7}

Table 3: Numerically computed coefficients $c_{\ell m}$ up to $\ell = 2$.

Summing the squared moduli for $\ell = 2$,

$$P\left(E = \frac{3\hbar^2}{m\alpha^2}\right) = \sum_{m=-2}^2 |c_{2m}|^2 = 0.19371.$$

Probability of Measuring $L_z = 2\hbar$

Since $L_z Y_{\ell}^m = m\hbar Y_{\ell}^m$, measuring $L_z = 2\hbar$ requires $m = 2$. Truncating at $\ell = 10$, the relevant coefficients are:

ℓ	m	$\text{Re}(c_{\ell m})$	$\text{Im}(c_{\ell m})$
3	-3	2.40×10^{-6}	-0.126487
3	-2	-0.221348	-5.00×10^{-7}
3	-1	4.49×10^{-7}	0.0390271
3	0	-0.223947	0
3	1	-4.49×10^{-7}	0.0390271
3	2	-0.221348	5.00×10^{-7}
3	3	2.40×10^{-6}	0.126487
4	-4	0.0735408	-7.09×10^{-7}
4	-3	-3.11×10^{-2}	-0.164052
4	-2	-0.0899834	-2.03×10^{-7}
4	-1	-1.77×10^{-6}	-0.154379
4	0	-0.142051	0
4	1	1.77×10^{-6}	-0.154379
4	2	-0.0899834	2.03×10^{-7}
4	3	-3.11×10^{-2}	-0.164052
4	4	0.0735408	7.09×10^{-7}

 Table 4: Coefficients $c_{\ell m}$ for $\ell = 3$ and $\ell = 4$.

The approximate probability is

$$P(L_z = 2\hbar) \approx \sum_{\ell=2}^{10} |c_{\ell,2}|^2 = 0.12021.$$

Time Evolution of $H(\theta, \phi, t)$ on the Sphere

The time evolution is reconstructed by expanding H in the eigenbasis and applying the temporal phase factors:

$$H(\theta, \phi, t) = \sum_{\ell m} c_{\ell m} Y_{\ell}^m(\theta, \phi) e^{-iE_{\ell}t/\hbar}.$$

The probability density $|H(\theta, \phi, t)|^2$ is simulated up to $\ell = 10$ and animated on the spherical surface. The animation is available at the following link: [Click here](#).

Discussion

This project consolidated several key concepts in quantum mechanics. The spherical harmonics $Y_{\ell}^m(\theta, \phi)$, rather than being abstract special functions, emerge here as the natural basis for any angular distribution on the sphere — whether a prescribed function like $g(\theta, \phi)$,

a binary-valued surface like the tennis ball $T(\theta, \phi)$, or a Gaussian wave packet evolving in time.

The expansion of $g(\theta, \phi)$ illustrated how even a relatively complex angular function can be captured by a small number of modes, with the dominant contributions concentrated at $\ell = 2$, $m = \pm 1$. The tennis ball analysis provided a striking counterexample: a binary, discontinuous function requires many high-order modes and exhibits clear Gibbs-type overshoot at the seam, making explicit the trade-off between truncation order and reconstruction fidelity.

For the free particle on the sphere, the connection between the angular momentum operator and the Hamiltonian was established rigorously, yielding eigenenergies $E_\ell = \hbar^2 \ell(\ell+1)/(2m\alpha^2)$ directly from the spectrum of L^2 . The numerical computation of measurement probabilities — $P(E = 3\hbar^2/m\alpha^2) \approx 0.194$ and $P(L_z = 2\hbar) \approx 0.120$ for the Gaussian initial state — demonstrated how the expansion coefficients translate directly into physical predictions.

References

1. Gutiérrez-Vega, J. C. (2025). *Notes for the Quantum Mechanics Course: Central Forces and Spherical Harmonics*. Tecnológico de Monterrey.
2. Gutiérrez-Vega, J. C. (2025). *Spherical Harmonics: Definitions, Properties and Examples*. Tecnológico de Monterrey.

Appendix: Source Code

Spherical Harmonic Routine and Expansion of $g(\theta, \phi)$

Core routine: `rutina_armonicosesfericos`

```
1 function Y = rutina_armonicosesfericos(l, m, theta, phi)
2 % Evaluates the normalized spherical harmonic Y_l^m(theta, phi)
3 % over arbitrary-shaped grids of theta and phi.
4
5 if l < 0 || l ~= round(l)
6     error('l must be a non-negative integer.');
```

```
7 end
8 if abs(m) > 1 || m ~= round(m)
9     error('|m| must be <= 1 and m must be an integer.');
```

```
10 end
11 if ~isequal(size(theta), size(phi))
12     error('theta and phi must have the same size.');
```

```
13 end
14
15 sz = size(theta);
16 th = theta(:).'; % 1 x N row vector
17 z = cos(th);
18
```

```

19     Pl_all = legendre(l, z);
20
21     mm = abs(m);
22     Pl_vec = squeeze(Pl_all(mm+1, :));
23     Pl = reshape(Pl_vec, sz);
24
25     N = sqrt((2*l + 1)/(4*pi) * factorial(l-mm)/factorial(l+mm));
26
27     phase = exp(1i * m .* phi);
28
29     Y = N * Pl .* phase;
30 end

```

Routine test: $(\ell, m) = (4, 3)$

```

1 clear; clc; close all;
2
3 l = 4; m = 3;
4 Ntheta = 100; Nphi = 200;
5
6 theta_vec = linspace(0, pi, Ntheta);
7 phi_vec = linspace(0, 2*pi, Nphi);
8 [TH, PH] = meshgrid(theta_vec, phi_vec);
9
10 Y = rutina_armonicosesfericos(l, m, TH, PH);
11
12 % Cartesian coordinates for surface plot
13 X = sin(TH) .* cos(PH);
14 Yc = sin(TH) .* sin(PH);
15 Z = cos(TH);
16
17 figure;
18 surf(X, Yc, Z, real(Y));
19 shading interp; axis equal off; colorbar;
20 title('Real part of Y_4^3');
21
22 figure;
23 surf(X, Yc, Z, imag(Y));
24 shading interp; axis equal off; colorbar;
25 title('Imaginary part of Y_4^3');

```

Normalization constant and expansion coefficients $C_{\ell m}$

```

1 clear; clc; close all;
2
3 %% Normalization constant A
4 Ntheta_fine = 5000;

```

```

5 theta_fine = linspace(0, pi, Ntheta_fine);
6 f_theta = theta_fine.^2 .* (pi - theta_fine).^4 .* sin(theta_fine);
7 I_theta = trapz(theta_fine, f_theta);
8 A = ((5*pi/8) * I_theta)^(-1/2);
9
10 %% Angular grid
11 Ntheta = 150; Nphi = 200;
12 theta = linspace(0, pi, Ntheta);
13 phi = linspace(0, 2*pi, Nphi);
14 [TH, PH] = meshgrid(theta, phi);
15 dtheta = theta(2) - theta(1);
16 dphi = phi(2) - phi(1);
17
18 %% Evaluate g
19 g = A .* TH .* (pi - TH).^2 .* cos(PH - 2*TH).^3;
20
21 %% Compute expansion coefficients
22 lmax = 15;
23 C_lm = zeros(lmax+1, 2*lmax+1);
24
25 for l = 0:lmax
26     for m = -l:l
27         Ylm = rutina_armonicosesfericos(l, m, TH, PH);
28         integrand = g .* conj(Ylm) .* sin(TH);
29         C_lm(l+1, m + lmax + 1) = sum(integrand, 'all') * dtheta * dphi;
30     end
31 end

```

Reconstruction and error analysis

```

1 %% Truncated reconstruction
2 Frec = zeros(size(TH));
3 for l = 0:lmax
4     for m = -l:l
5         Ylm = rutina_armonicosesfericos(l, m, TH, PH);
6         Frec = Frec + C_lm(l+1, m + lmax + 1) .* Ylm;
7     end
8 end
9
10 %% Error analysis
11 g_abs = abs(g);
12 abs_err = abs(Frec - g);
13
14 eps0 = 1e-8 * max(g_abs, [], 'all');
15 mask_rel = g_abs > eps0;
16
17 rel_err = zeros(size(g));

```



```

18 rel_err(mask_rel) = abs_err(mask_rel) ./ g_abs(mask_rel);
19
20 max_rel_err = max(rel_err(mask_rel));
21 max_abs_zero = max(abs_err(~mask_rel), [], 'all');
22
23 fprintf('Maximum relative error = %.4e\n', max_rel_err);
24 fprintf('Maximum absolute error (g approx 0) = %.4e\n', max_abs_zero);

```

Hammer Projection, Spin States, and Tennis Ball

Hammer projection and spherical harmonic plotting

```

1 % Hammer projection utilities and problem set solutions
2
3 A = 0.148;
4 a = 0.01;
5 theta = 0:a:pi;
6 phi = -pi:a:pi;
7 [T, P] = meshgrid(theta, phi);
8
9 % Problem 1: g(theta,phi) in Hammer projection
10 g = @(T,P) A .* T .* (cos(P - 2.*T)).^3 .* (pi - T).^2;
11 G = g(T, P);
12 Hammer(G, T, P, 'g(\theta,\phi) in Hammer projection');
13
14 % Problem 2b: Y_6^9 in Hammer projection
15 [Xm,Ym,Zm,Ylm] = Arm_Esf(6, 9, T, P, true);
16 Hammer(real(Ylm), T, P, 'Re(Y_9^6(\theta,\phi)) in Hammer');
17
18 % Problem 2c: Spin function G_9^6 in Hammer projection
19 G_s = Spin(6, 9, T, P);
20 Hammer(real(G_s), T, P, 'Re(G_9^6(\theta,\phi)) in Hammer');
21
22 % Problem 2d: Probability density |G_{-3/2}^{5/2}|^2 in Hammer
23 G_s2 = Spin(-3/2, 5/2, T, P);
24 Hammer(abs(G_s2).^2, T, P, '|G_{5/2}^{-3/2}(\theta,\phi)|^2 in Hammer');
25
26 %% Helper functions
27 function [X,Y] = Hammer(G, theta, phi, tit)
28     X = (sqrt(8).*sin(theta).*sin(phi./2))./sqrt(1 + sin(theta).*cos(phi./2));
29     Y = (sqrt(2).*cos(theta))./sqrt(1 + sin(theta).*cos(phi./2));
30     surf(X, Y, G);
31     shading interp;
32     xlabel('X'); ylabel('Y');
33     title(tit); colorbar;
34 end
35

```

```

36 function [Xm,Ym,Zm,Ylm] = Arm_Esf(m, l, theta, phi, graf)
37     Plm = legendre(l, cos(theta));
38     if l ~= 0
39         Plm = reshape(Plm(m+1,:,:), size(phi));
40     end
41     C = sqrt((2*l+1)*factorial(l-m) / (4*pi*factorial(l+m)));
42     Ylm = C .* Plm .* exp(1i*m*phi);
43     [Xm,Ym,Zm] = sph2cart(phi, pi/2-theta, abs(real(Ylm)));
44     if graf
45         surf(Xm,Ym,Zm,real(Ylm)); shading interp;
46         title("Spherical harmonic m=" + m + ", l=" + l); colorbar;
47     end
48 end
49
50 function G = Spin(m, s, theta, phi)
51     a = sqrt((2*s+1)*factorial(2*s) / (4*pi*factorial(s+m)*factorial(s-m)));
52     G = a .* cos(theta./2).^(s+m) .* sin(theta./2).^(s-m) .* exp(1i*m*phi);
53 end

```

Tennis ball model and harmonic expansion

```

1  %% Tennis ball construction
2  dth = 0.01;
3  theta_vec = 0:dth:pi;
4  phi_vec = -pi:dth:pi;
5  [TH, PH] = meshgrid(theta_vec, phi_vec);
6
7  Delta_theta = pi/6;
8  theta_seam = pi/2 + Delta_theta * sin(2*PH);
9
10 Tfun = ones(size(TH));
11 Tfun(TH < theta_seam) = -1;
12
13 % Cartesian and Hammer coordinates
14 Xes = sin(TH).*cos(PH);
15 Yes = sin(TH).*sin(PH);
16 Zes = cos(TH);
17
18 Xh = (sqrt(8)*sin(TH).*sin(PH/2))./sqrt(1 + sin(TH).*cos(PH/2));
19 Yh = (sqrt(2)*cos(TH))./sqrt(1 + sin(TH).*cos(PH/2));
20
21 %% Expansion coefficients C_lm
22 clear; clc; close all;
23 dth = 0.01;
24 theta_vec = 0:dth:pi; phi_vec = -pi:dth:pi;
25 [TH, PH] = meshgrid(theta_vec, phi_vec);
26 Delta_theta = pi/6;

```

```

27 theta_seam = pi/2 + Delta_theta * sin(2*PH);
28 Tfun = ones(size(TH));
29 Tfun(TH < theta_seam) = -1;
30
31 dtheta = theta_vec(2) - theta_vec(1);
32 dphi = phi_vec(2) - phi_vec(1);
33 lmax = 15;
34 C = zeros(lmax+1, 2*lmax+1);
35
36 for ell = 0:lmax
37     for m = -ell:ell
38         Ylm = rutina_armonicosesfericos(ell, m, TH, PH);
39         integrand = Tfun .* conj(Ylm) .* sin(TH);
40         C(ell+1, m + lmax + 1) = sum(integrand, 'all') * dtheta * dphi;
41     end
42 end
43
44 %% Truncated reconstruction
45 Frec = zeros(size(TH));
46 for ell = 0:lmax
47     for m = -ell:ell
48         Ylm = rutina_armonicosesfericos(ell, m, TH, PH);
49         Frec = Frec + C(ell+1, m + lmax + 1) * Ylm;
50     end
51 end
52
53 figure;
54 surf(Xes, Yes, Zes, real(Frec));
55 shading interp; axis equal off; colorbar;
56 title('Truncated reconstruction F(\theta,\phi)');

```

Free Particle on the Sphere

```

1 a = 0.01;
2 theta = 0:a:pi;
3 phi = -pi:a:pi;
4 [T, P] = meshgrid(theta, phi);
5 dtheta = theta(2) - theta(1);
6 dphi = phi(2) - phi(1);
7
8 %% Normalized initial state
9 H0 = exp(-((T - pi/3)/(pi/12)).^2) .* exp(-(P - pi/2)/(pi/6)).^2);
10 integral_H2 = sum(sum(abs(H0).^2 .* sin(T))) * dtheta * dphi;
11 Cnorm = 1 / sqrt(integral_H2);
12 H = Cnorm * H0;
13
14 %% Expansion coefficients c_{lm}

```

```

15 lmax = 10;
16 c = zeros(lmax+1, 2*lmax+1);
17
18 for l = 0:lmax
19     for m = -l:l
20         if m < 0
21             [~,~,~,Ypos] = Arm_Esf(abs(m), l, T, P, false);
22             Ylm = (-1)^m * conj(Ypos);
23         else
24             [~,~,~,Ylm] = Arm_Esf(m, l, T, P, false);
25         end
26         integrand = conj(Ylm) .* H .* sin(T);
27         c(l+1, m + lmax + 1) = sum(sum(integrand)) * dtheta * dphi;
28     end
29 end
30
31 %% Measurement probabilities
32 P_E = sum(abs(c(2+1, (-2:2) + lmax + 1)).^2);
33 fprintf('P(E = 3 hbar^2/m alpha^2) = %.5f\n', P_E);
34
35 idx_m2 = 2 + lmax + 1;
36 P_Lz = sum(abs(c(3:lmax+1, idx_m2)).^2);
37 fprintf('P(L_z = 2 hbar) = %.5f\n', P_Lz);
38
39 %% Time evolution
40 tmax = 4; Nt = 120;
41 tvec = linspace(0, tmax, Nt);
42 [Xs, Ys, Zs] = sph2cart(P, pi/2 - T, 1);
43
44 figure;
45 axis equal off; colormap turbo;
46
47 for it = 1:Nt
48     t = tvec(it);
49     Ht = zeros(size(T));
50     for l = 0:lmax
51         El = l*(l+1);
52         for m = -l:l
53             if m < 0
54                 [~,~,~,Ypos] = Arm_Esf(abs(m), l, T, P, false);
55                 Ylm = (-1)^m * conj(Ypos);
56             else
57                 [~,~,~,Ylm] = Arm_Esf(m, l, T, P, false);
58             end
59             Ht = Ht + c(l+1, m + lmax + 1) .* exp(-1i*El*t) .* Ylm;
60         end
61     end
62     surf(Xs, Ys, Zs, abs(Ht).^2);

```

```
63 shading interp; axis equal off;  
64 title(sprintf('Time evolution: t = %.2f', t));  
65 colorbar; drawnow;  
66 end
```